

Day 3: Introduction to IP Addressing Theory :IPv4 and IPv6: Differences, structure, and importance. Subnetting basics: CIDR, subnet masks .Practical :Assign static IPs to devices and verify connectivity. Task: Solve subnetting exercises and calculate valid subnets.

Tasks:-

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Ans. :-

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Address format: 192.168.1.1 (decimal).	Address format: 2001:db8::1 (hexadecimal).
Supports about 4.3 billion unique addresses.	Supports 340 undecillion (3.4×10^{38}) unique addresses.
Uses NAT (Network Address Translation) because of limited address space.	Usually does not require NAT due to the vast address space.
Header is more complex and has variable length.	Header is simpler and has a fixed length for faster routing.
Broadcast communication is supported.	Broadcast is not supported; it uses multicast and anycast instead.
IP configuration is mainly manual or via DHCP.	Supports SLAAC (Stateless Address Autoconfiguration) and DHCPv6.
Security (IPsec) is optional.	IPsec support is built into the IPv6 standard.

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Ans.:-

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Ans.:-

Calculate the Number of Subnets and Hosts :-

- **Network:** 192.168.1.0/24
- **New Subnet Mask:** /27 (255.255.255.224)

Number of subnets:

- Borrowed bits = $27 - 24 = 3$
- Number of subnets = $2^3 = 8$

Hosts per subnet:

- Host bits = $32 - 27 = 5$
- Total IP addresses per subnet = $2^5 = 32$
- Usable hosts = $32 - 2 = 30$ (excluding the network and broadcast addresses)

Answer:

- Valid Subnets: 8
- Usable Hosts per Subnet: 30

First Three Subnets:-

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0/27	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95

**4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.**

Ans.:-

Given:

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

Find the Network Address:-

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0
- Network Address: 10.0.5.0

Find the Broadcast Address:-

For a /24 network, the last address in the subnet is the broadcast address:

- Broadcast Address: 10.0.5.255

Find the Valid Host Range:-

- First Valid Host: 10.0.5.1
- Last Valid Host: 10.0.5.254

Item	Value
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR Notation	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254

5. Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Ans.:-

IPv4 Subnet Calculator

Result

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	00001010000000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

Network Class ☒ Any ☐ A ☐ B ☐ C

Subnet

IP Address

My manual calculation matched the subnet calculator's output: the network address is 10.0.5.0, the broadcast address is 10.0.5.255, and the valid host range is 10.0.5.1–10.0.5.254.